27/8/2026 Thursday

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Case study - 1.

✱ BrightMart ✱

is a medium sized online shopping company, customers use it's website and mobile application to purchase products.

The company ~~source~~ stores :-

i) The customer names
ii) E-mail addresses
iii) Phone no.
iv) Delivery addresses.
v) Passwords
vi) order information
vii) Payment related information

BrightMart as around 100 Employees.

The company recently had new employee To make things easier, the IT department give employees ~~across~~ access to different internal systems.

12 YEARS OF ESTABLISHMENT
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However, the company has the following security weakness:-

- Some employees use simple passwords such as BrightMart1234.
- Many employees use same passwords for multiple systems.
- Software updates are sometimes delayed.
- The company does not regularly conduct employees cybersecurity training.
- There is no multi-factor authentication for some important accounts.
- Employees can access sensitive information that they do not necessarily need for their jobs.
- Backups are performed only once a month.

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The incident

One Monday morning an employee named Rahul receives an E-mail.

Subject : Urgent - Your accounts will be suspended.

The emails appears to comes from the company's IT department

It says :-

" Your company account will be Suspended today. Click the link below and verify your passwords immediately."

Rahul clicks the link,

The page looks almost identical to the comany login page.
He enters the username and passwords
Rahul does not realize that whole website is controlled by the attacker
The attacker now ar credinals of rahul.

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Stage 1:- Unauthorized Access.

The attacker use rahul's credintals. to access BrightMart's internal system.

Because, Rahul uses the same passwords, For another systems, the attacker is also able to access that system.

The attacker discovers that Rahul had access to customer information.
The attacker downloads a large amounts of customer files.

What ~~type of~~ cybersecurity problem has occured?

Stage 2:- Malware Attack.

The attacker ~~stands~~ sends email to rahul constaining Malicious link.

The malware begins communicating with the attacker's system
The attacker now attempts to move from Rahul's computer to other system with BrainMarth's network.
which vulnerability allow him to do?

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Stage 3:- Ransomware

Two days later, employees arrives at work and discover that they cannot access important files.

A message appears on their computers:-

" Your files have been encrypted. Pay the ransom to recover them?

The company's customer databases and several important business files are encrypted Unfortunately, the company's most recent backup is almost a ~~20~~ month ~~did~~ old

BrightMart's website stops working.

Customers cannot place orders

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1) What cybersecurity problem has occured?
=) phishing, credential theft, Unauthorize access, data breach.

2) Which vulnerability allowed the attacker to move further into the organization?
=) Weak passwords, Excessive user privileage lack of MFA, Poor network security, lack of employee awareness, outdated software.

3) What threats are present in this case?
=) Phishing, malware, Randsomware, Credentia theft, Unauthorized access.

4) What weakness in BrightMart's securit allowed the attack to succeed?
=)
Weak password -> gave acess
password reuse -> weak password
NO MFA -> No authentication easy log
Lack of training -> less awarness

5) Was confidential information exposed?
=) Yes
customer information was accessed by th attacker

6) Was information changed or damaged?
=) Yes
The attacker could modify or encrypt file

7) Were systems available to legitimate users?
2) No.

Confidentiality → Affected
Integrity → Affected
Availaibility → Affected.

8) What should BrightMart do to prevent this from happening again?
2)
Strong passwords.
MFA
Keep software updated.
Maintain backups.
Spread awareness.

9) Who is responsible for cyber security?
2) IT Employees, & every employees also there mistakes

10) Was phishing the only problem?
2) No,
There are many other problems poor backups, No, MFA and etc.

11) What was the biggest Vulnerability?
2) There is no single correct answer Students should justify there choice.

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5) "Lack of employee awareness was a major vulnerability because the attack started when rahul clicked the phising link".

Quick Assement.

1) Initial attack on BrightMart
=> Data Theft

2) Give three vulnerability ?
=> No MFA, weak password, phishing

3) Which three security principles of CIA Triad affected?
=> Confidentiality, Integerity & availability.

4) Impact of ransomware?
=> Data Theft

5) Three Security controls implement?
=> stay update, spread awarness, Strong password.

Rahul 3/4/26